# C. V. Raman Global University
# Odisha

R24

Registration No. : | 2 | 3 | 0 | 1 | 0 | 2 | 0 | 0 | 2 | 2 |

*Total number of printed pages – 3*

B. Tech.
**Subject Code: MA203**

**3RD SEMESTER REGULAR EXAMINATION – 2024**

**DISCRETE MATHEMATICS AND GRAPH THEORY**

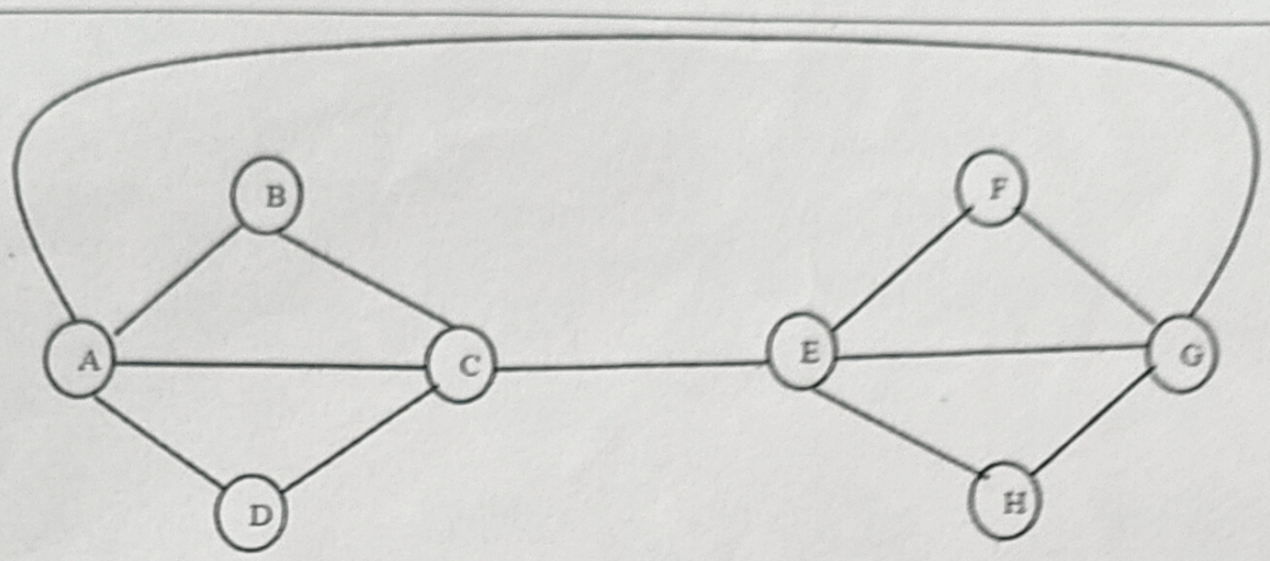
**BRANCH: CSE AND ALL ALLIED BRANCHES**

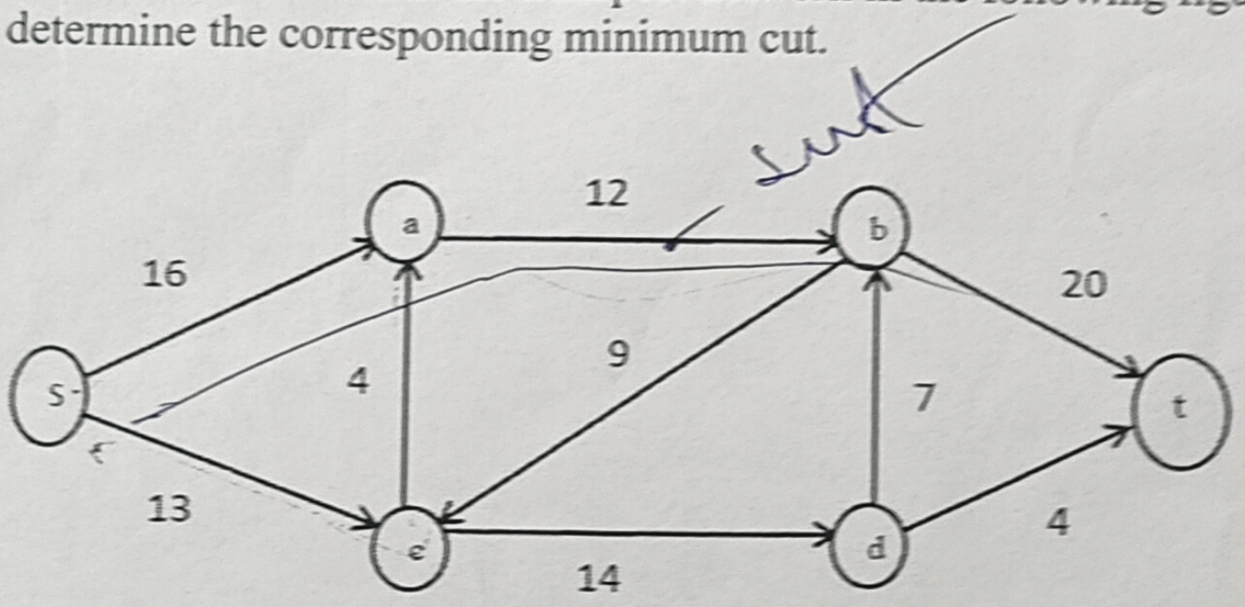
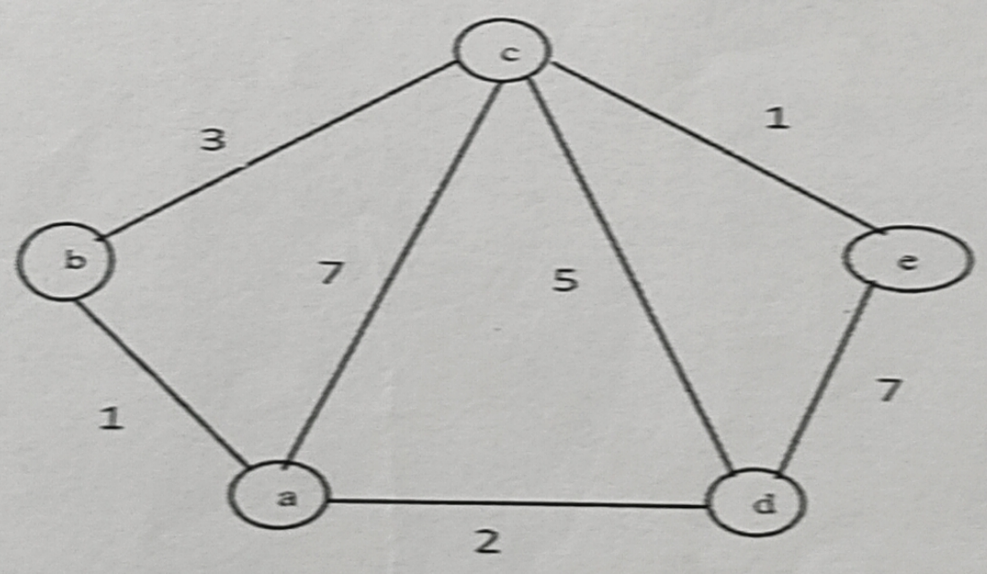
**Time – 3 Hours**

**Full Marks – 100**

Answer all Questions from Part – A and Part – B

| Q1. | Part-A (ANSWER ALL QUESTIONS) | CO | Marks | BL |
|---|---|---|---|---|
| (a) | Show that $P \vee \sim P$ is a tautology. | CO1 | 2 | 2 |
| (b) | State De Morgan's law for quantifier. | CO1 | 2 | 1 |
| (c) | Find the maximal and minimal elements of the partially ordered set $(\{1,2,3,4,6,8\},/)$ | CO1 | 2 | 2 |
| (d) | How many solutions does the equation $x + y + z = 17$ have, where $x, y, z$ are non-negative integers ? | CO2 | 2 | 1 |
| (e) | Find the generating function for the sequence 0,1,2,3,4…. | CO2 | 2 | 2 |
| (f) | Evaluate the particular solution of the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 7^n$ for all $n \geq 2$. | CO2 | 2 | 2 |
| (g) | If 6 colours are used to paint 37 home.Show that at least 7 home of them will be of same colour. | CO2 | 2 | 2 |
| (h) | A graph contain 21 edges, 3 vertices of degree 4 and all other vertices of degree 2. Calculate the total number of vertices in the graph. | CO3 | 2 | 1 |
| (i) | How many edges does $K_{2,6}$ contain? | CO3 | 2 | 1 |
| (j) | Find the chromatic number of complete graph$K_5$. | CO3 | 2 | 2 |
| (k) | Define complete binary tree. | CO4 | 2 | 1 |
| (l) | Find all the spanning trees of complete graph $K_4$. | CO4 | 2 | 2 |




| | | | | |
|---|---|---|---|---|
| **Q5.** | **Answer the followings:** | | | |
| (a) | Find the maximum flow in the transport network in the following figure and determine the corresponding minimum cut.   | CO4 | 7 | 3 |
| (b) | Find the shortest distance and shortest path from 'a' to 'e' in the given weighted graph using Dijkstra algorithm.   | CO4 | 7 | 3 |
| **Q6.** | **Answer the followings:** | | | |
| (a) | Show that the set of all positive rational numbers forms an abelian group under the composition define by $a * b = \frac{ab}{2}$ | CO5 | 7 | 2 |
| (b) | Determine the (2 ,5) group code function $e_H : B^2 \rightarrow B^5$ for the parity matrix H= $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | CO5 | 7 | 3 |

25.07.25 2nd Sit

# C. V. Raman Global University
## Odisha

25

Registration No. :

*Total number of printed pages – 3*

B.Tech
Subject Code: MA108

**3RD SEMESTER REGULAR EXAMINATION – 2025**

**DISCRETE MATHEMATICS**

**BRANCH: CSE and all Allied Branches**

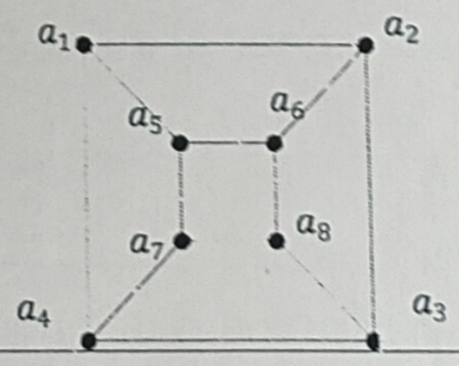
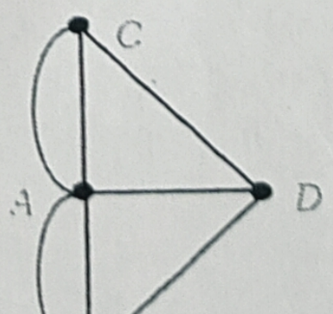
**Time – 3 Hours**

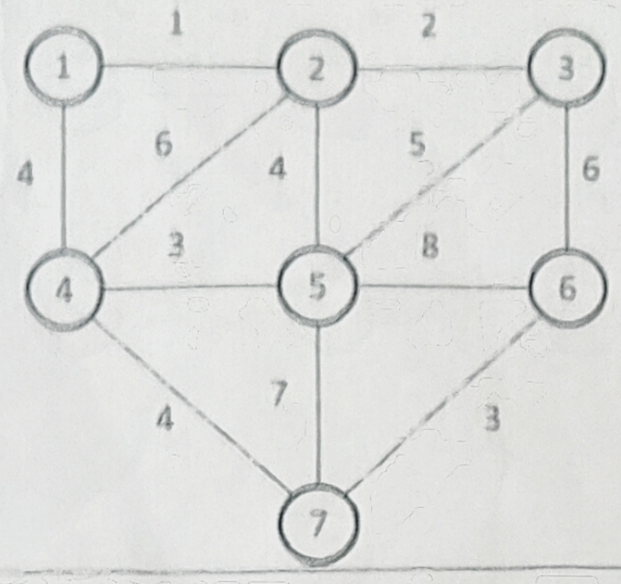
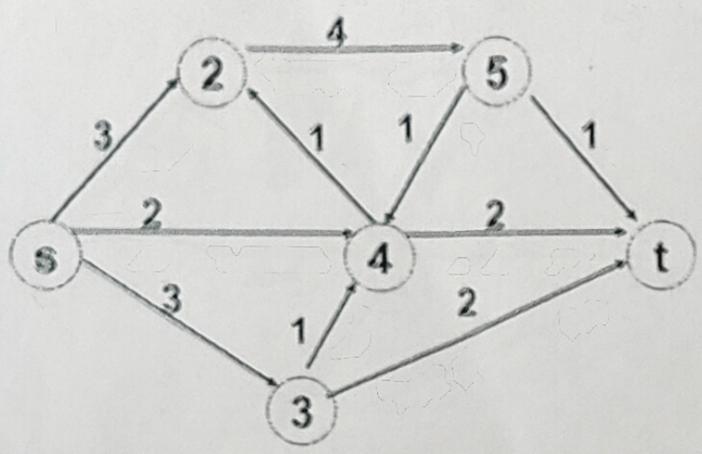
**Full Marks – 100**

Answer all Questions from Part – A and Part – B

| Q1. | Part-A (ANSWER ALL QUESTIONS) | CO | Marks | BL |
|---|---|---|---|---|
| (a) | Construct the truth table for the relation $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$. | CO1 | 2 | 3 |
| (b) | Use mathematical induction to prove that $n^2 + n$ is divisible by 2 for all n $\in \mathbb{N}$. | CO1 | 2 | 2 |
| (c) | Find the transitive closure of the relation R = {(1, 2), (2, 3)} on the set {1, 2, 3} | CO1 | 2 | 3 |
| (d) | How many solutions are there of $x + y + z = 17$ subject to the constraints $x \geq 1, y \geq 2$ and $z \geq 3$. | CO2 | 2 | 3 |
| (e) | Solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 2$. | CO2 | 2 | 2 |
| (f) | How many ways can 8 people be arranged in a line if two specific people must stand together? | CO2 | 2 | 3 |
| (g) | State Handshaking theorem. | CO3 | 2 | 1 |
| (h) | Write the Euler's formula for a planer graph with 4 components. | CO3 | 2 | 1 |
| (i) | What is the chromatic number of an odd cycle? | CO3 | 2 | 1 |
| (j) | Define spanning tree. Give an example. | CO4 | 2 | 1 |
| (k) | A tree has two vertices of degree 2 , one vertex of degree 3 and three vertices of degree 4. How many vertices of degree 1 does it have? | CO4 | 2 | 1 |
| (l) | State the Max-Flow Min-Cut Theorem for a network. | CO4 | 2 | 1 |
| (m) | Show that the set of integers does not form a group under multiplication. | CO5 | 2 | 3 |

| | | | | |
|---|---|---|---|---|
| (n) | Define a subgroup of a group. Give an example. | CO5 | 2 | 1 |
| (o) | Construct a binary code with a minimum Hamming distance of 2 | CO5 | 2 | 3 |

| | Part-B | CO | Marks | BL |
|---|---|---|---|---|
| **Q2.** | **Answer any two:** | | | |
| (a) | Prove using a truth table that $(P \vee Q) \wedge (\neg P \vee R)$ is logically equivalent to $(P \wedge R) \vee (Q \wedge \neg R)$. | CO1 | 7 | 2 |
| (b) | Apply Warshall's Algorithm to compute the transitive closure of a relation represented by the adjacency matrix below: $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ | CO1 | 7 | 2 |
| (c) | Define a poset. Draw the Hasse diagram for the divisors of 60 under the "divides" relation and verify whether that forms a poset or not. | CO1 | 7 | 1 |
| **Q3.** | **Answer any two:** | | | |
| (a) | Use generating function to solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $, a_0 = 1$ | CO2 | 7 | 2 |
| (b) | Solve $a_{n+2} - 4a_{n+1} + 4a_n = 2^n$ | CO2 | 7 | 3 |
| (c) | In how many ways can 5 people be seated in a row if two of them, Alice and Bob, must sit next to each other? | CO2 | 7 | 2 |
| **Q4.** | **Answer any two:** | | | |
| (a) | Define bipartite Graph. Check whether the following graph is bipartite or not?   | CO3 | 7 | 1 |
| (b) | Define Euler circuit. State the necessary and sufficient condition for a connected graph to have a Euler circuit.<br>Check whether the following graph has a Euler circuit or not.   | CO3 | 7 | 1 |
| (c) | State and prove Euler's formula for a connected planer simple graph | CO3 | 7 | |

| | | | | |
|---|---|---|---|---|
| **Q5.** | **Answer the followings:** | | | |
| (a) | Use Kruskal's algorithm find a minimal spanning tree (MST ) of the following graph.   | CO4 | 7 | 2 |
| (b) | Find the maximal flow of the following network using Ford-Filkerson algorithm   | CO4 | 7 | 3 |
| **Q6.** | **Answer the followings:** | | | |
| (a) | Prove that the fourth roots of unity $1, -1, i, -i$ form an abelian multiplicative group | CO5 | 7 | 3 |
| (b) | For the (2,4 ) encoding function determine how many errors can detect and how many errors can correct with respect to following data $e(00) = 0000, e(10) = 0110, e(01) = 1011, e(11) = 1100.$ | CO5 | 7 | 3 |

**Best of Luck**

C. V. Raman Global University Odisha

B25

Registration No. : | | | | | | | | | | |

*Total number of printed pages –04*

B.Tech.
MTH21105

3rd SEMESTER BACK EXAMINATION – 2025

**SUBJECT NAME: Discrete Mathematics**

**Branch(s): CSIT**

**Time – 3 Hours**
**Full Marks – 100**
Answer all Questions from Part – A and Part – B
The figures in the right hand margin indicate marks.

## PART – A

1. Answer the following questions: [15x 2]

(a) Show that q→(iPAQ) is a tautology.

(b) Find the maximal and minimal elements of the partially ordered set ({2,4,5,10,12,20,25}, |).

(c) Let n be an integer and 3n+2 be odd. Then show that n is odd by method of contrapositive.

(d) Find number of derangements of the set {1,2,3,4}.

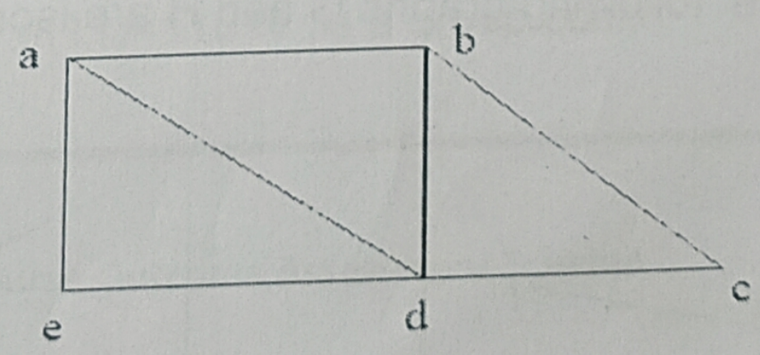
(e) Find the particular solution of the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$, for all $n \geq 2$.

(f) State Pigeonhole principle.

(g) Show that $K_5$ is not planar.

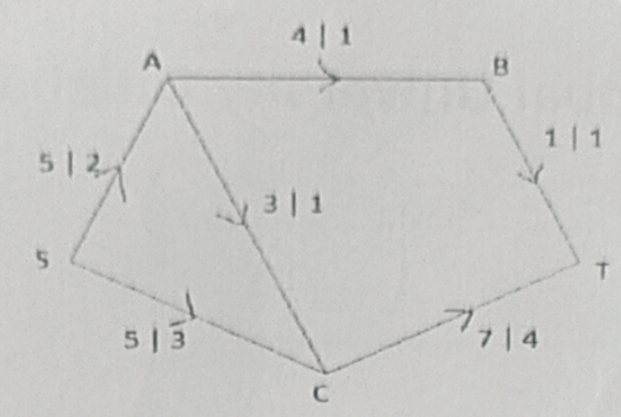
(h) How many edges does $K_{3,6}$ contain?

(i) Determine whether the following graph has Euler circuit or Euler path and find it.




(j) Define spanning tree.

(k) Check whether the chain SCABT flow augmented path or not?




(l) Find the generating function for the sequence 1,1,1,1, ....

(m) Find the distance between $x = 110110$, and $y = 010010$.

(n) Write all the subgroups of the group G= ($Z_6$, $+_6$).

(o) Find all the generators of the cyclic group G= ({1,3,7,9},$\times_{10}$).

## PART - B

2. Answer any two: [7 x 2]

(a) Show that the division relation on set of Integer is a partial ordering relation. Draw the Hasse diagram representing the partial ordering {(a, b)| a divides b} on {1, 2, 3, 4, 6, 8, 12}.

(b) Find the transitive closure of the relation $R = \{(1,2),(2,1),(2,3),(3,4)\}$ on the set $A = \{1,2,3,4\}$ using Warshall's algorithm.

(c) Prove that $n^3 - n$ is divisible by 3 for every positive integer $n$ by using mathematical induction.

3. Answer any two: [7 x 2]

(a) Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$, for all $n \geq 2$ by using generating function.

(b) How many integers between 1 and 500 (inclusive) are

(i) divisible be at least one of 3, 5, 7?

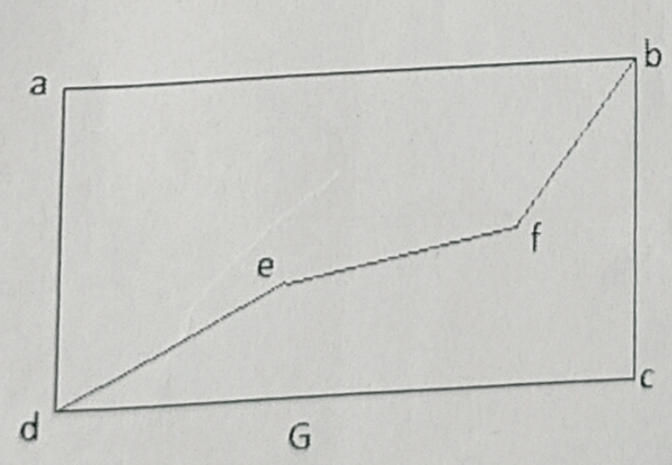
(ii) divisible by 3 and 5 but not by 7?

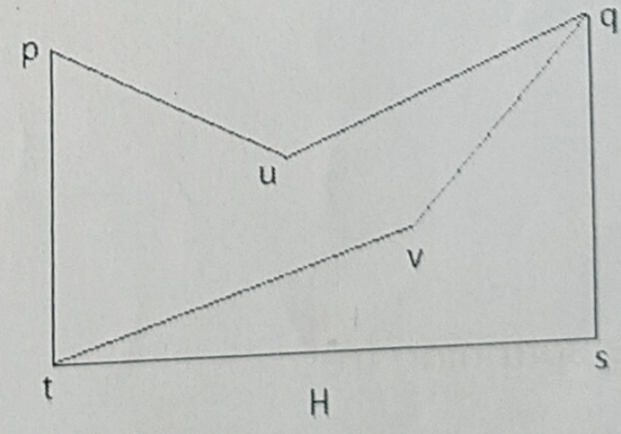
(c) Find the sequence $b_r$ from the generating function $\sum_{r=0}^{\infty} b_r x^r = \frac{1}{x^2-5x+6}$

4. Answer any two: [7 x 2]

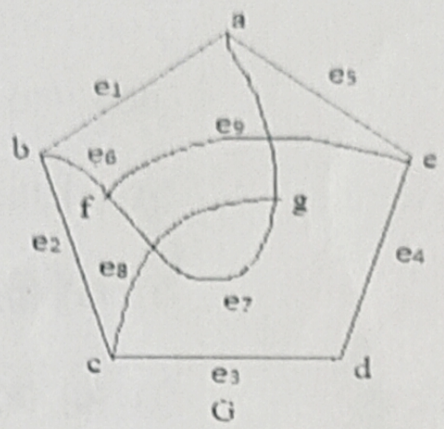
(a) Prove that the number of vertices with odd degree is always even in a simple graph.

(b) Test whether the following graphs G and H are isomorphic or not .



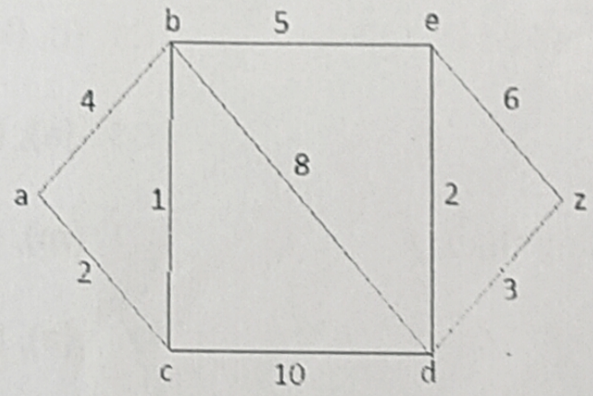
(c) Define adjacency matrix and incidence matrix and also find the adjacency and incidence matrices of the following graph G.



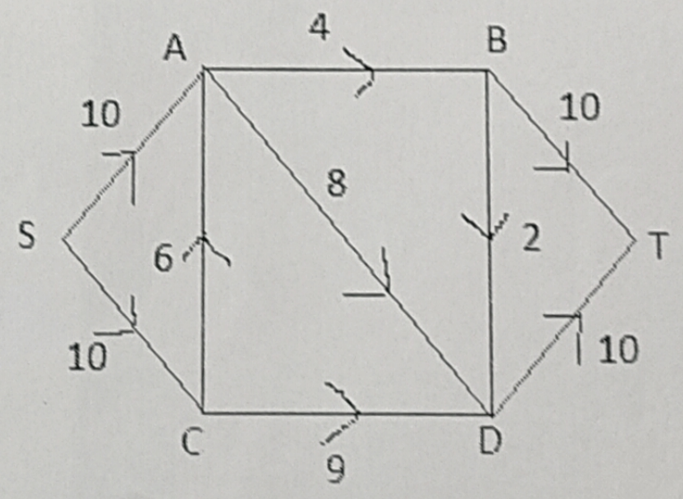
G

5. Answer the following: [7 x 2]

(a) Find the length of a shortest path between the vertices a and z by using Dijkstra's algorithm in the following graph




(b) Find the maximum flow of the following network using Ford-Fulkerson algorithm.




6. Answer the following: [7 x 2]

(a) Check whether the algebraic structure $G = (\left\{\begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in Q\right\}; *)$, where $A * B = A.B$ forms a group or not?

(b) Show that (2,5) encoding function defined by e(00)=00000, e(01)=01110,e(10)=10101 ,e(11)=11011 ,is a group code.

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**Course Outcome Assessment Scheme**

25

# C. V. Raman Global University
## Odisha

Registration No. :

*Total number of printed pages – 3*

B.Tech
Subject Code: MA203

**3RD SEMESTER SUMMER QUARTER EXAMINATION – 2025**

**SUBJECT NAME: DISCRETE MATHEMATICS AND GRAPH THEORY**

**BRANCH:CSE and all Allied Branches**

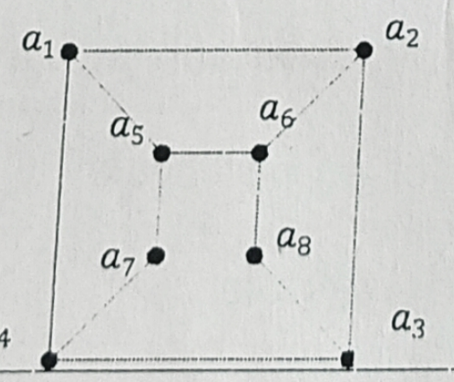
**Time – 3 Hours**

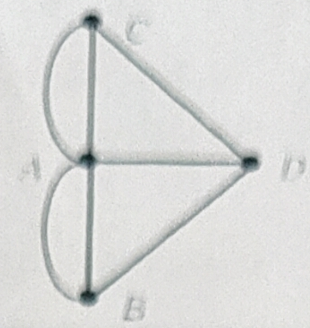
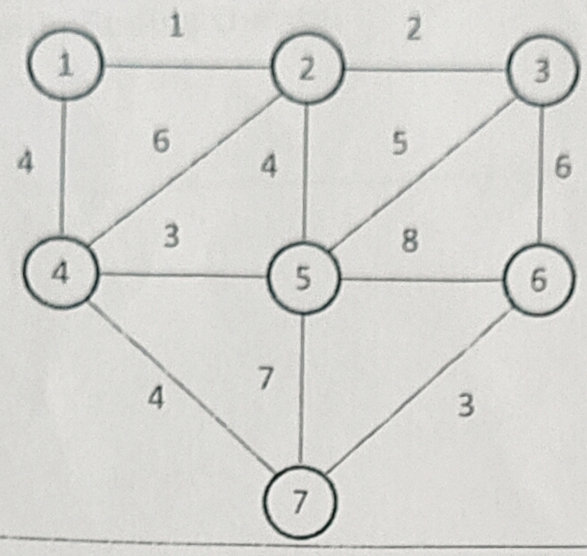
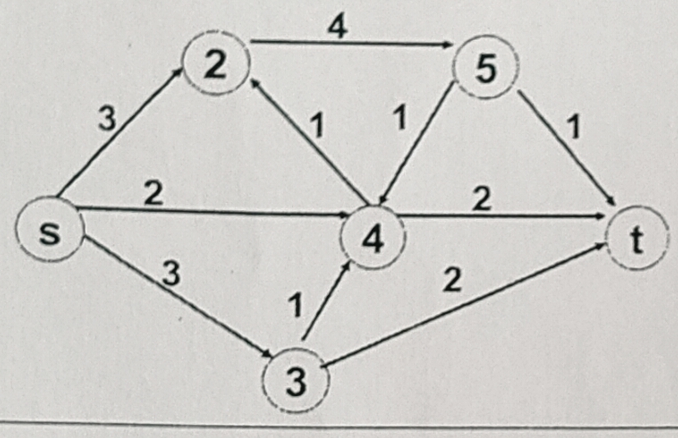
**Full Marks – 100**

Answer all Questions from Part – A and Part – B

| Q1. | Part-A (ANSWER ALL QUESTIONS) | CO | Marks | BL |
|---|---|---|---|---|
| (a) | Construct the truth table for the relation $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$. | CO1 | 2 | 3 |
| (b) | Use mathematical induction to prove that $n^2 + n$ is divisible by 2 for all $n \in \mathbb{N}$. | CO1 | 2 | 2 |
| (c) | Find the transitive closure of the relation R = {(1, 2), (2, 3)} on the set {1, 2, 3} | CO1 | 2 | 3 |
| (d) | What is the maximum number of groups into which you can distribute 30 students such that at least one group must have at least 5 students? | CO2 | 2 | |
| (e) | Solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 2$. | CO2 | 2 | |
| (f) | How many ways can 8 people be arranged in a line if two specific people must stand together? | CO2 | 2 | |
| (g) | State Handshaking theorem. | CO3 | 2 | |
| (h) | Write the Euler's formula for a planer graph with 4 components. | CO3 | 2 | |
| (i) | What is the chromatic number of an odd cycle? | CO3 | 2 | |
| (j) | Define spanning tree. Give an example. | CO4 | 2 | |
| (k) | If T is a binary of height 8, then what will be the maximum number of terminal vertex for T? | CO4 | 2 | |
| (l) | State the Max-Flow Min-Cut Theorem for a network. | CO4 | 2 | |

| | | | | |
|---|---|---|---|---|
| (m) | Show that the set of integers does not form a group under multiplication. | CO5 | 2 | 3 |
| (n) | Define a subgroup of a group. Give an example. | CO5 | 2 | 1 |
| (o) | Construct a binary code with a minimum Hamming distance of 2 | CO5 | 2 | 3 |

| | Part-B | CO | Marks | BL |
|---|---|---|---|---|
| **Q2.** | **Answer any two:** | | | |
| (a) | Prove using a truth table that $(P \vee Q) \wedge (\neg P \vee R)$ is logically equivalent to $(P \wedge R) \vee (Q \wedge \neg R)$. | CO1 | 7 | 2 |
| (b) | Apply Warshall's Algorithm to compute the transitive closure of a relation represented by the adjacency matrix below: $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ | CO1 | 7 | 2 |
| (c) | Define a poset. Draw the Hasse diagram for the divisors of 60 under the "divides" relation and verify whether that forms a poset or not. | CO1 | 7 | 1 |
| **Q3.** | **Answer any two:** | | | |
| (a) | Use generating function to solve the recurrence relation $a_r - 3a_{r-1} + 2a_{r-2} = 0$ for $r$=2,3,4... and initial condition $a_0 = 2$, and $a_1 = 3$ | CO2 | 7 | 2 |
| (b) | What is the solution of the recurrence relation $a_n = a_{n-1} + a_{n-2} + 3n + 1 \; for \; n \geq 2,$ with $a_0 = 2$ and $a_1 = 3$? | CO2 | 7 | 3 |
| (c) | In how many ways can 5 people be seated in a row if two of them, Alice and Bob, must sit next to each other? | CO2 | 7 | |
| **Q4.** | **Answer any two:** | | | |
| (a) | Define bipartite Graph. Check whether the following graph is bipartite or not?   | CO3 | 7 | 1 |
| (b) | Define Euler circuit. State the necessary and sufficient condition for a connected graph to have a Euler circuit. | CO3 | 7 | 1 |

| | Questions | Marks | Total Mark |
|---|---|---|---|
| | Check whether the following graph has a Euler circuit or not.   | | |
| (c) | Prove that every planar graph has a vertex of degree at most 5. | CO3 | 7 | 2 |
| **Q5.** | **Answer the followings:** | | | |
| (a) | Use Kruskal's algorithm find a minimal spanning tree (MST ) of the following graph.   | CO4 | 7 | 2 |
| (b) | Find the maximal flow of the following network using Ford-Filkerson algorithm   | CO4 | 7 | 3 |
| **Q6.** | **Answer the followings:** | | | |
| (a) | State Lagrange's Theorem.<br>If G is a group of order 10 and H is a subgroup of order 5, how many distinct left cosets does H have in G? | CO5 | 7 | 1 |
| (b) | Construct a binary linear code using a generator matrix and explain how syndrome decoding can be used to correct errors. | CO5 | 7 | 3 |

**Best of Luck**